

- 3 (a) The collared flycatcher, *Ficedula albicollis*, and the pied flycatcher, *F. hypoleuca* are two closely related species of bird. DNA analysis has shown that speciation from a common ancestor occurred approximately 1 million years ago.

A study was carried out on the island of Öland, Sweden. In Öland, the breeding areas of the two bird species overlap and small numbers of hybrid flycatchers are produced.

- Birds were captured and their DNA was analysed to identify whether each bird was *F. albicollis*, *F. hypoleuca* or a hybrid.
- Sperm samples were taken from the male birds.

Table 3.1 shows the percentage of males of each bird type with normal sperm.

Table 3.1

bird type	percentage of males with normal sperm
<i>F. albicollis</i>	68
<i>F. hypoleuca</i>	78
male hybrid	0

- The researchers observed that female birds mostly choose mates of their own species based on plumage (feathers) and song.
- Hybrid flycatchers are produced when female *F. albicollis* mate with male *F. hypoleuca* that have a song that is similar to *F. albicollis*.
- Analysis showed that all female hybrids were sterile.

The group of eggs a female bird lays at a single time in its nest is called a clutch. The offspring in the nest are looked after by a male-female pair. Sometimes the male in the male-female pair does not provide the sperm that fertilise the eggs of the female.

Table 3.2 shows:

- the percentage of clutches with eggs that hatched
- the percentage of extra-pair nestlings (offspring in the nest fathered by a male that was different from the male of the male-female pair).

Table 3.2

parents of nest		percentage of clutches with eggs that hatched	percentage of extra-pair nestlings
male	female		
<i>F. albicollis</i>	<i>F. albicollis</i>	94.5	17.2
<i>F. hypoleuca</i>	<i>F. hypoleuca</i>	89.3	22.4
hybrid	<i>F. albicollis</i> or <i>F. hypoleuca</i>	38.0	100.0

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- (b) A single-nucleotide polymorphism (SNP) is caused by a base pair substitution mutation in a specific region of DNA.

One method of identifying whether two individuals have the same SNP is to:

- use a specific primer and polymerase chain reaction (PCR)
- add a restriction enzyme
- carry out gel electrophoresis
- stain with a dye to compare banding patterns.

(i) Explain why:

a specific primer is used

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a restriction enzyme is added

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gel electrophoresis is carried out.

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- (ii) The method of identifying whether two individuals have the same SNP method was carried out to compare species **A** and species **B**.

Fig. 3.1 shows the banding patterns that were observed.

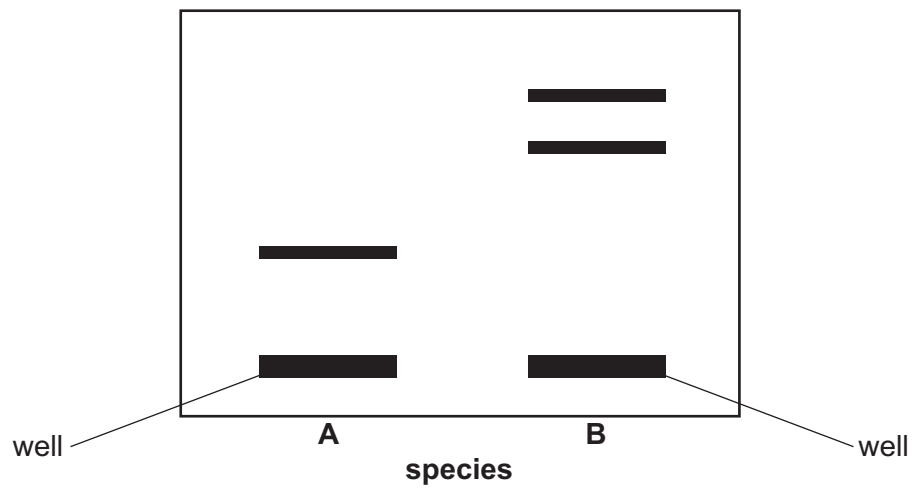


Fig. 3.1

Describe **and** suggest an explanation for the results obtained in Fig. 3.1.

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